Writing a Physics Experimental Report
Step-by-Step Checklist

1 Aim



What are you trying to find out or prove?

- ☐ Clear, specific aim linked to the experiment.
- ☐ Mentions the key variables or the quantity to be determined.
- ☐ Written in one or two sentences.

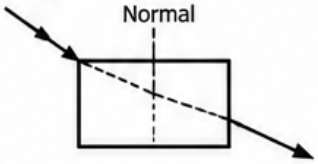
2 Equipment



What did you use?

- ☐ All main equipment listed.
- ☐ Named clearly (e.g. glass block, laser, protractor).
- ☐ Include measuring instruments.

3 Diagram



What does your set-up look like?

- ☐ Clear, neat labelled diagram of the experimental set-up.
- ☐ Includes key features (e.g. incident and refracted rays, normal, equipment).
- ☐ Labels are clear and correct.

4 Variables

Independent	
Dependent	
Controlled	

What are you changing, measuring and keeping the same?

- ☐ Independent variable identified and described.
- ☐ Dependent variable identified and described.
- ☐ Controlled variables listed.
- ☐ Brief note on how each is treated.

5 Method



How did you do it?

- ☐ Written in clear, logical steps (numbered).
- ☐ Includes key details (e.g. how to measure, repeat readings).
- ☐ Can be followed by someone else.

6 Precautions



How did you get reliable and safe results?

- ☐ At least 3–5 relevant precautions.
- ☐ Linked to reducing specific errors or risks.
- ☐ Clear, practical statements (not just "be careful").

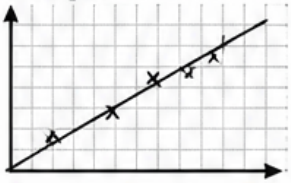
7 Results (Data)

Reading no.	Variable 1 (units)	Variable 2 (units)	Calculated values (if any)
1			
2			
3			

What did you measure and/or calculate?

- ☐ Results table with clear headings and units.
- ☐ All readings recorded to a suitable number of decimal places.
- ☐ Calculated values included (if required, e.g. sin i and sin r).

8 Graph



SALT check:

- **S** – Scale (suitable and even)
- **A** – Axes (correct variables)
- **L** – Labels (with units)
- **T** – Title (states what is plotted)

- ☐ Correct variables on each axis (independent on x-axis).
- ☐ Suitable scale, using most of the graph paper.
- ☐ Axes labelled with quantity and unit.
- ☐ Clear, correct title.
- ☐ Points plotted accurately with a line of best fit.

9 Analysis / Calculations



What result(s) did you obtain from your graph or data?

- ☐ Required calculations shown clearly (e.g. gradient, mean).
- ☐ Units included.
- ☐ Final value(s) stated with appropriate number of significant figures.

10 Conclusion, Evaluation and Improvements

PEER conclusion	Limitations (Cause → Consequence)	Improvements
P – Point **E** – Evidence **E** – Explanation **R** – Reflection	e.g. wide ray → uncertain ray path → larger uncertainty in angles	e.g. use a narrower ray to give a sharper ray path

- ☐ Clear conclusion linked to the aim (use PEER).
- ☐ States whether the results support the model/relationship.
- ☐ References data or graph.
- ☐ Identifies at least 2 limitations (cause and consequence).
- ☐ Suggests realistic improvements.